

AtmosFlow AI Chat — Technical Audit

In-app "Field Assistant" agent (codename Jasper) · codebase: atmosiq-v6 · live at atmosflow.net
Prepared for Prudence Safety & Environmental Consulting, LLC · June 2026 · Model audited: claude-sonnet-4-6

The AtmosFlow AI chat is the in-app Field Assistant, codenamed Jasper — a streaming, tool-using Anthropic Claude agent scoped as a "Defensibility Copilot." It explains, retrieves, drafts, and screens, but is explicitly forbidden from making final calls (scoring, compliance, causation). That boundary is the product's legal moat.

1. Capabilities

Conversational features

- **Token streaming** over Server-Sent Events (SSE) for chat-quality UX.
- **Persistent, resumable conversations** stored in Postgres; survive across sessions, with auto-generated titles.
- **Context-aware:** each request carries an AssessmentContext block (current zone, logger data, instruments) grounding answers in the live walkthrough.
- **Structured four-section answer format** for context-bearing questions: Assessment context !' Screening interpretation !' Recommended next steps !' Defensibility note — always closing with the literal line "IH Review Required."
- **Thumbs-up/down feedback** per assistant message; conversation history browsing.

Tool surface (7 tools)

Tier	Tool	Function
L2	lookup_exposure_limit	OSHA PEL / NIOSH REL / ACGIH TLV / EPA NAAQS, primary-source cited
L2	lookup_sampling_method	NIOSH NMAM / OSHA / EPA TO sampling methods
L2	lookup_health_effects	ATSDR / IARC / EPA IRIS health endpoints
L2	list_known_analytes	Discovery / fallback over the knowledge base
L3	search_standards_corpus	RAG over curated standards corpus (TF-IDF + cosine)
L4	analyze_photo	Multimodal Claude-vision screening of attached field photos
—	propose_action	Agentic: proposes navigate / add_zone_note as an Accept/Reject card; never executes server-side
—	generate_report	Proposes a DOCX render from the user's saved templates (token-fill only)

Retrieval, vision & guardrails

- **RAG:** pure-JS TF-IDF + cosine similarity over ~40 curated standards chunks (ASHRAE 62.1/55/241, OSHA Z-tables, NIOSH, EPA NAAQS, IICRC S520). Deliberately not embeddings — chosen for zero external deps, determinism (auditable), and sub-ms latency.
- **Vision:** same claude-sonnet-4-6 model, strict screening-only JSON schema, always stamped ih_review_required: true; optional focus modes (mold / moisture / hvac / ventilation / dust / general).
- **Hard guardrails (role prompt):** may not assign scores/severity, determine compliance, diagnose health effects, attribute causation, certify safe/unsafe, override the engine/calibration gate, mutate records, or invent any value/citation/serial. Tool output is always preferred over recalled values.

2. Engineering build

Runtime & shape

- **Vercel serverless function** (api/field-assistant.ts, ~1,040 lines, Node 20) — production deploy path.
- **Direct fetch to the Anthropic Messages API** (no SDK). Model claude-sonnet-4-6, anthropic-version 2023-06-01, temperature 0.7, max_tokens 800.
- **Hand-rolled SSE parser** demuxes Anthropic content-block frames and re-emits its own events: meta, token, tool_start, tool_call, proposed_action, render_proposed, done, error.
- **Agentic loop** capped at MAX_TOOL_ROUNDS = 4, with a graceful "tool budget reached" note on overrun.

Cost & performance engineering

- **Prompt caching:** role prompt + standards + FAQ sent as cache_control: ephemeral system blocks (~25–35K tokens) so the cached prefix bills at ~10% after warm-up; dynamic context kept last to preserve the cached prefix.
- **Explicit cost model** !' per-turn estimated_cost_usd (verified accurate — see §4).
- **Bounded:** 20-turn history cap, 4,000-char message cap, 5 photos/request, ~2 MB/photo.

Data & infrastructure

- **Auth:** Supabase bearer token. Persistence: field_assistant_conversations + field_assistant_messages (Postgres); telemetry columns added in migration 015.
- **Rate limiting:** 15/min burst, 150/day, 10/day on free tier; counted against the narrative_generations ledger; plus an UNLIMITED_USAGE_EMAILS allowlist.
- **Privacy:** persisted copies run through scrubPii; the in-flight copy to Anthropic stays raw.
- **Observability:** audit-log rows jasper_asked / photo_analyzed; every implicit-500 path carries a stable diagnostic code (fa_init_000...fa_init_005).

Two hard-won architectural rules (documented prod incidents): (1) Lean hot path — generate_report returns a render proposal; the actual docxtemplater render runs in a separate endpoint so heavy deps never load on a Jasper cold-start. (2) No extension-less .ts imports from .js files on the api/** graph — once crashed every Jasper turn in production while passing all local gates.

3. Harnesses

A. Test harnesses (Vitest, ~20 files touching Jasper)

- **field-assistant.test.ts** — handler (auth, rate limit, SSE, loop).
- **field-assistant-bundle.test.ts** — esbuild bundle assertion: fails if docxtemplater/pizzip/renderTemplate appear (enforces lean hot path), with a positive control on the render endpoint.
- **field-assistant-prompt.test.ts** — pins the role-prompt guardrails.
- **Tool & component tests** — field-assistant-tools, generate-report-tool, propose-action-dispatcher, buildJasperContext; FieldAssistant-photos, jasper-feedback-row, jasper-message-actions, JasperWatchPanel.
- **Injection pattern** — handlers export __test hooks (setSupabase, setFetch) rather than relying on vi.mock.

B. Guardrail / CI harnesses

- **check-api-js-imports.mjs** — walks the import graph from api/** and fails on extension-less .ts imports from .js; wired into npm run lint:imports and acceptance criterion API-JS-IMPORT-GUARDRAIL.
- **Acceptance harness** — acceptance-check.mjs + JSON gate configs (prod-ready 23 / pricing-rollout 19 / go-live 21). Tasks aren't "done" until the relevant group exits 0.

C. Runtime & data harnesses

- **Express container harness** (server/index.js) — parallel deploy path to Vercel serverless, for FedRAMP/GovCloud container portability.
- **Fine-tune export harness** (scripts/export-finetune-dataset.mjs) — exports persisted conversations (with telemetry) into a claude-sonnet-4-6 fine-tuning dataset shape, suggesting a planned model-improvement loop.

4. Cost model verification

The handler hardcodes the Anthropic price list for claude-sonnet-4-6 and computes per-turn estimated_cost_usd. All four constants were checked against Anthropic’s current published rates:

Constant	Code value	Official Sonnet 4.6 rate	Match
COST_INPUT_PER_M	\$3.00 /M	\$3.00 /M (list input)	'
COST_OUTPUT_PER_M	\$15.00 /M	\$15.00 /M (list output)	'
COST_CACHE_READ_PER_M	\$0.30 /M	\$0.30 /M (0.1x input)	'
COST_CACHE_WRITE_PER_M	\$3.75 /M	\$3.75 /M (1.25x input, 5-min TTL)	'

No drift — constants are accurate as of June 2026. Caveat: \$3.75 assumes the 5-minute (default) cache TTL the code actually uses; a 1-hour TTL would bill 2x input = \$6.00/M. Constants are hardcoded and won’t auto-update if Anthropic changes rates.

5. Assessment summary

This is a genuinely well-engineered, production-grade agent, not a thin chatbot wrapper. Standouts: the screening-only / engine-sacred guardrail design, prompt-cache cost engineering, the deterministic TF-IDF RAG choice (auditable by design), the propose-action human-in-the-loop pattern, and unusually disciplined regression harnesses born from real prod incidents.

Observations (design notes, not bugs):

- **TF-IDF retrieval is lexical-only** — misses semantic matches where query/corpus vocab don’t overlap (embeddings are the planned L3.5).
- **No SDK / hardcoded anthropic-version** — keeps the function lean but means API-contract changes are a manual bump.
- **Rate limits piggyback on narrative_generations** — convenient, but couples chat quota to the narrative ledger.